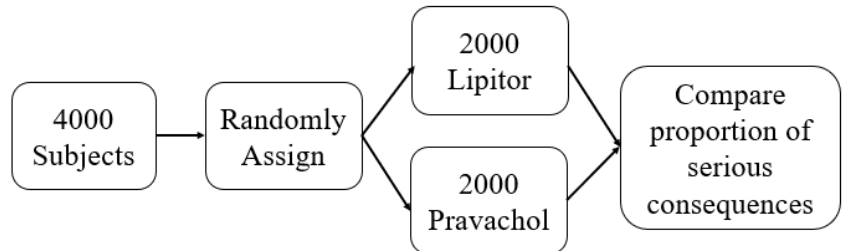


## Why Do We Have This?

- Creating sampling distribution of a difference in sample proportions will set us up for success when we study inferences in Unit 6.
- These sampling distributions can be used to answer the common question of “which is better?” For example:
  - Which of two popular drugs – Lipitor or Pravachol – helps lower “bad cholesterol” more?
  - Researchers used 4000 people with heart disease as subjects in a completely randomized experiment. They were randomly assigned to one of two treatment groups: Lipitor or Pravachol.
  - At the end of the study, researchers compared the proportion of subjects in each group who had died, had a heart attack, or suffered other serious consequences within two years.
  - For the subjects assigned to Pravachol, 0.263 suffered a serious consequence.
  - For the subjects assigned to Lipitor, 0.224 suffered a serious consequence.



Does this mean that Lipitor is better at lowering bad cholesterol? Is this difference a result of Lipitor being better or is it merely due to chance involved in the random assignment of treatments?

Answers to these questions require a sampling distribution of a difference in sample proportions!

## Recall: Sampling Distribution of $\hat{p}$

| Shape         | Center        | Spread        |
|---------------|---------------|---------------|
|               |               |               |
| As long as... | As long as... | As long as... |

# Recall: Combining Random Variables

- Shape: Combining two or more normal distributions will result in a combined normal distribution.
- Center:  $\mu_{X-Y} = \mu_X - \mu_Y$
- Spread:  $\sigma_{X-Y} = \sqrt{\sigma_X^2 + \sigma_Y^2}$

## Sampling Distribution of $\hat{p}_1 - \hat{p}_2$

\_\_\_\_\_ = sample proportion from the first group                      \_\_\_\_\_ = sample proportion from the second group

| Shape         | Center         | Spread        |
|---------------|----------------|---------------|
|               |                |               |
| As long as... | As long as.... | As long as... |

To answer the original question, if there is a difference between Lipitor and Pravachol, we need to know a little more about hypothesis testing. For now, let’s look at an example involving finding probabilities of a difference between population proportions.

Example) A recent study examined the percentage of young adult women and men who get less than 7 hours of sleep a night. In this study, it was found that 45% of women and 38% of men get less than 7 hours of sleep a night. What is the probability that a random sample of 50 women and 75 men will result in a difference of 15% or more between the two groups? *(There are two probabilities implied here!)*